Logical framework and activity marix

	Results chain	***Indicator***	***Source and mean of verification***	***Assumptions***
Impact (Overall objective)	*Improve environmental health and living conditions in Bissau through a sustainable, efficient and socially inclusive solid waste management system*	*Quantitative:* *• Reduction of unmanaged waste in public areas by 30–40% within 3 years* *Increase in households with access to regular waste collection from baseline to 70%.* *Qualitative:* *• At least 70% of surveyed households report improved cleanliness and living conditions.*	• *Health statistics from local clinics/Ministry of Health.* • *Household surveys and community perception surveys.*	*Not applicable*
Outcome (Specific objective)	*Establish a Sustainable solid waste management system*	*Quantitative*: *• At least 80% of neighborhoods receive waste collection at the planned frequency • At least 85% of waste collection vehicles remain operational at any given time.* *Qualitative*: *• 70% of users and 75% of staff report satisfaction with service reliability.*	• *SWM unit administrative reports (staffing, performance).* • *Annual municipal budget and expenditure reports.* • *Activity monitoring database.*	• *Stable security and socio-economic conditions in Bissau* • *Continuous political support from municipal authorities*

Outputs				***Assumptions for each outputs:***
	1. Improve waste infrastructure and collection system	*1.Quantitative:* *• At least 20 functioning collection points, 3–* *4 transfer areas, and 90% of planned collection routes operating as scheduled.* *Qualitative:* *• 70% of residents rate accessibility and cleanliness of collection points as "good" or "very good".*	***Common** sources of verification for **all outputs** :* • *SWM unit organizational structure documentation* • *Workshop reports and feedback forms* • *Attendance lists from events.* • *Social media analytics (views, shares)* • *School participation reports* • *Municipal financial reports and updated budgets* • *Site rehabilitation reports and photos* • *Medical check-up records from clinics*	*1. Communities accept the installation of new waste points*
	2. Strength institutional and technical capacity	*2.Quantitative:* *• 100 staff trained, with 60–70% applying new operational procedures after 6 months.* *Qualitative:* *• Managers report at least a 30% improvement in institutional performance and problem-solving capacity.*		2. *Institutional reforms supported by municipality leadership ; Low staff turnover during the project period*
	3. Enhance community participation and environmental awareness	*3.Quantitative:* *• Minimum 25 awareness activities conducted. + 50% of community members demonstrate correct waste practices (sorting, disposal, engagement).* *Qualitative:* *• 60% of community members show positive attitudes toward waste sorting and municipal services.*		3. *Communities are willing to participate in awareness activities ;* *Communication channels (radio, social media, events) remain accessible*
	4. Improve financial management and	*4.Quantitative:* *• 3 financial tools (budget templates,*		4. *Municipal authorities commit*

	partnership for SWM 5. *Improve the working condition (health and safe) for waste workers*	*cost-tracking tools, tariffs) updated. + 5 partnership agreements operational.* *Qualitative:* *• At least 70% of stakeholders report improved financial transparency and partnership effectiveness.* *5.Quantitative:* *• 90% of workers equipped with appropriate* *PPE. + Reduction of workplace safety incidents by 40%.* *Qualitative:* *• 75% of workers report improved safety,* *dignity and comfort at work.*		*to adopting financial reforms；* 5. *Workers are willing to participate in training and health check-ups；* *Waste management companies cooperate with safety requirements*

Activity Matrix

1. Improve waste infrastructure and collection system： *1.1 survey and assess existing waste collection points and transfer stations* *1.1.1 visit sites, check equipment conditions, identify*	***Means*** *What are the political, technical, financial, human and material resources required to implement these activities, e. g. staff, equipment, supplies, operational facilities, etc.*	***Assumptions*** 1. *Communities accept the installation of new waste points：* *1.1*

priority areas *1.2 Rehabilitate and upgrade damaged collection points and small transfer areas.* *1.2.1 Hire local contractors and supervise construction work.* *1.3 Build new waste collection points in unserved neighborhoods.* *1.3.1 Prepare simple designs and obtain local permits.* *1.4. Strengthen waste collection equipment and ensure its proper management* *1.4.1. Procure essential waste collection equipment (trucks, bins, tools) and replenish stocks* *1.4.2. Establish a maintenance and repair system for equipment and facilities* *1.4.3. Set up a monitoring system for equipment condition and availability* *1.4.4. Plan and optimize collection routes to ensure regular coverage across all neighborhoods*	***Costs*** *What are the action costs? How are they classified? (Breakdown in the Budget for the Action)*	
2. Strength institutional and technical capacity *2.1. Conduct structured capacity-building programs for local institutions* *2.1.1. Assess existing technical and institutional*		

capacity gaps *2.1.2. Train municipal staff on technical solid waste management operations* *2.1.3. Train staff on environmental and health aspects* *2.2. Develop and implement standard operating procedures (SPO), for waste collection and transport* *2.2.1. Draft and validate SOPs through workshops with municipal staff* *2.2.2. Train field teams and supervisors on the use of the SOPs* *2.2.3. Pilot the SOPs in selected neighborhoods and adjust based on feedback* *2.3. Establish a dedicated SWM unit* *2.3.1. Define the unit's structure, roles, staffing needs, and mandate* *2.3.2. Assign or recruit staff and provide basic operational training* *2.3.3. Equip the unit with essential office and communication tools.* *2.4. provide technical equipment and tools* *2.4.1. Identify priority equipment needs based on*		

operational gaps *2.4.2. Procure and distribute equipment to relevant teams* *2.5. facilitate knowledge exchange and partnership* *2.5.1. Organize peer-learning workshops with other municipalities or agencies* *2.5.2. Establish collaborations with NGOs, universities, and regional SWM networks*		
3. Enhance community participation and environmental awareness *3.1 Establish a communication and community outreach team* *3.1.1. Recruit or designate a media/outreach team responsible for awareness activities* *3.2. Raise public awareness through targeted communication* *3.2.1. Produce and disseminate educational materials on social media and in public spaces* *(short videos, posters, infographics about good waste practices)* *3.2.2. Integrate waste management awareness sessions into school programs*		

(basic environmental education modules, interactive activities) *3.2.3. Organize community events to promote proper waste management* *(clean-up days, demonstration sessions, information booths)* *3.3. Engage key influencers and community leaders* *3.3.1. Train religious and traditional leaders to promote responsible waste practices* *3.4 Explain to the population the goal of the project*		
4. Improve financial management and partnership for SWM *4.1 Review and update the city waste management budget.* *4.2 Train municipal staff in basic accounting and cost tracking.* *4.3 Create a simple fee system to help cover waste service costs.* *4.4 Develop public–private partnership (PPP) agreements with local companies.* *4.5 Prepare short financial reports and share results with the public.*		
5. Improve the working condition (health and safe) for waste workers		

5.1 Do more control on the respect of the work conditions in the companies *5.2 Provide protective equipment and safety kits to waste workers.* *5.3 Conduct health and safety training on waste handling and hygiene.* *5.4 Rehabilitate sorting and collection sites with safe and clean facilities.* *5.5 Organize regular medical check-ups for waste workers.* *5.6 Formalize informal workers through cooperatives and fair contracts.*		